



Different interventions in preventing opioid-induced cough: a meta-analysis[☆]

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Abstract

Background: Cough is one of the most common complications of opioids. Many studies have evaluated the effect of various drugs in preventing opioid-induced cough (OIC). However, there is existing controversy about those reports. The present study was performed to assess the efficacy of different interventions on OIC.

Methods: We searched the Cochrane Central Register of Controlled Trials (CENTRAL), PubMed, and Embase to identify randomized controlled trials on the efficacy of different drugs in the prevention of OIC. Opioids included fentanyl, sufentanil, and remifentanyl. We mainly investigated the incidence and severity of OIC after different interventions.

Results: Thirty-four trials including 9906 patients were analyzed in this study. Twenty different drugs were reported, and 10 drugs were identified in more than 2 articles. These drugs, including lidocaine, ketamine, dexmedetomidine, priming of fentanyl, propofol, dezocine, dexamethasone, dextromethorphan, and magnesium sulfate (MgSO₄), showed a significant efficacy compared with controls. There were insufficient numbers of trials for salbutamol, clonidine, tramadol, pentazocine, rocuronium, midazolam, atropine, terbutaline, sodium chromoglycate, beclomethasone, and ephedrine. From these data, we found that salbutamol, tramadol, midazolam, and atropine were ineffective.

Conclusions: This meta-analysis suggested that the prophylactic administration of lidocaine, ketamine, dexmedetomidine, priming of fentanyl, propofol, and dezocine was effective in preventing OIC.

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1. Introduction

Opioids, such as fentanyl, sufentanil, and remifentanyl, are widely used in the induction and maintenance of general

anesthesia. Opioids are commonly applied due to their strong analgesia, rapid onset, short duration, and reduction of cardiovascular response. However, the administration of a bolus of opioids frequently elicits cough, with an incidence up to 65% [1]. This complication is called opioid-induced cough (OIC).

Most OIC is transient, light, and self-limiting. It is rarely noticed, although there is a high incidence. However, OIC is not always benign in all patients. It is widely known that cough is associated with increased intracranial, intraocular, and intra-abdominal pressure [2]. Moreover, severe OIC may result in

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multiple conjunctival and periorbital petechiae [3], or it may even cause upper airway obstruction [4]. This is very dangerous in patients with coexisting diseases, including increased intracranial pressure, brain hernia, cerebral aneurysm, brain trauma, increased ocular pressure, open eye injury, arterial aneurysm resection, full stomach, pneumothorax, or hypersensitive airway disease [5–7]. Therefore, it is essential to take effective interventions to prevent OIC in the clinical setting.

Numerous studies have reported various ways to manage OIC: slowing the injection rate, diluting drug concentration, reducing drug dose, using a peripheral injection site, verifying appropriate administration order of drugs, and instructing patients on the huffing maneuver [8–12]. However, some of these OIC management strategies are not effective enough or are limited in clinical applications. Pharmacological interventions have been widely researched to treat OIC. This systematic review sought to evaluate the effectiveness of those available drugs.

2. Methods

2.1. Search strategy and study selection

We conducted a systematic search of the PubMed, Embase, and CENTRAL databases (to December 2015) for randomized controlled trials (RCTs). Relevant studies were associated by comparing the efficacy of pharmacological interventions with controls in the prevention of OIC. In addition, the retrieved articles listed in the references were also reviewed. The search used the following keywords: opioid, fentanyl, remifentanyl, sufentanyl, and cough. Studies that met the following inclusion criteria were included: prospective RCTs published in English and comparison of prophylactic pharmacological interventions with controls on the incidence rate or severity.

2.2. Data extraction

Two authors independently reviewed the titles and abstracts to select the potential relevant studies and subsequently screened the full-text articles in detail. Disagreements were resolved through discussion or a third investigator. Data were extracted separately from studies by the 2 investigators to a single table (Table 1). The extracted data were as follows: first author, year of publication, Jadad score, interventions, type and dose of opioids, and patients' conditions. If the study included a combination of 2 drugs, we did not extract the data to analyze the combined intervention.

2.3. Study quality assessment

We used the modified 7-point Jadad scale to assess the quality of RCTs. The scale included randomization, allocation concealment, blinding method, and the description of dropouts [13,14]. In this scale, 0 to 3 points indicate a low quality, and 4 to 7 points indicate a high quality.

Table 1 The characteristics of the included 34 studies

Study	Jadad score	Intervention	Opioid	Patients
Bang et al, 2010 [15]	5	Saline Lidocaine 0.5 mg/kg Salbutamol inhaled	Remifentanyl: TCI 5 ng/mL (<1 min)	ASA I/ II aged 18-65 y
Guler et al, 2010 [16]	6	Lidocaine 1 mg/kg Ketamine 0.5 mg/kg Saline	Fentanyl: 1.5 µg/kg (>2 s) after 1 min	ASA I/ II aged 18-65 y
Kim et al, 2008 [41]	5	Lidocaine 0.5 mg/kg Saline	Remifentanyl: TCI 4 ng/mL after 1 min	ASA I/ II aged 18-65 y
Pandey et al, 2005 [28]	5	Lidocaine 0.5 mg/kg Lidocaine 1.0 mg/kg Lidocaine 1.5 mg/kg Placebo	Fentanyl: 3 µg/kg after 1 min	ASA I/ II aged 18-60 y
Lin et al, 2004 [29]	5	Saline Lidocaine 2 mg/kg Propofol 0.6 mg/kg Ephedrine 5 mg	Fentanyl: 2.5 µg/kg (<2 s) after 1 min	ASA I/ II aged 18-65 y
Pandey et al, 2004 [30]	5	Lidocaine 1.5 mg/kg Saline	Fentanyl: 3 µg/kg after 1 min	ASA I/ II aged 18-60 y
Saleh et al, 2014 [5]	5	Saline Ketamine 0.15 mg/kg Dexmedetomidine 0.5 mg/kg Ket 0.15 mg/kg + Dex 0.5 mg/kg	Fentanyl: 3 µg/kg (<5 s) after 1 min	ASA I/ II aged 18-70 y
Kim et al, 2009 [20]	6	Ketamine 0.1 mg/kg Saline	Remifentanyl: TCI 5 ng/mL (>5 s) after 1 min	ASA I/ II aged 18-65 y
Yeh et al, 2007 [37]	5	Ketamine 0.15 mg/kg Saline	Fentanyl: 1.5 µg/kg (<5 s) after 1 min	ASA I/ II aged 18-65 y
Honarmand et al, 2013 [44]	5	Ketamine 10 mg Dexamethasone 10 mg Ketamine 10 mg + Dex 10 mg Saline	Remifentanyl: 1 µg/kg (>10 s) after 5 min	ASA I/ II aged 18-70 y
He et al, 2012 [17]	5	Saline Dexmedetomidine 0.5 mg/kg Dexmedetomidine 1 mg/kg	Fentanyl: 4 µg/kg (<2 s) after 10 min	ASA I/ II aged 18-60 y
Sun et al, 2013 [25]	5	Saline Dexmedetomidine 0.1 µg/kg	Sufentanyl: 0.5 µg/kg (>3 s)	ASA I/ II aged 18-58 y

(continued on next page)

Table 1 (continued)

Study	Jadad score	Intervention	Opioid	Patients
Yu et al, 2012 [36]	5	Dexmedetomidine 0.25 µg/kg		
		Dexmedetomidine 0.5 µg/kg		
		Saline	Fentanyl: 3 µg/kg (<2 s) after 2 min	ASA I/II aged 18-65 y
		Midazolam 0.06 mg/kg		
Gu et al, 2012 [23]	6	Dexmedetomidine 0.6 µg/kg		
		Mid 0.06 mg/kg + Dex 0.6 µg/kg		
		Saline	Fentanyl: 2.5 µg/kg	ASA I/II aged 22-70 y
		Fentanyl 0.5 µg/kg	Fentanyl: 2 µg/kg	
Hung et al, 2010 [42]	5	Fentanyl 1 µg/kg	Fentanyl: 1.5 µg/kg	
		Fentanyl 1.5 µg/kg	Fentanyl: 1 µg/kg (<5 s) after 1 min	
		Saline	Fentanyl: 150 µg	ASA I/II aged 18-75 y
		Fentanyl 25 µg	Fentanyl: 125 µg	
Jung et al, 2011 [22]	5	Fentanyl 25 µg	Fentanyl: 150 µg after 1 min	
		Placebo	Fentanyl: 2 µg/kg	ASA I/II aged 18-75 y
		Fentanyl 0.5 µg/kg	Fentanyl: 1.5 µg/kg after 1 min, 2 min, 3 min (3-5 s)	
Shrestha et al, 2012 [10]	4	Saline	Fentanyl: 150 µg	ASA I/II aged 18-75 y
		Fentanyl 25 µg	Fentanyl: 125 µg	
		Fentanyl 25 µg	Fentanyl: 150 µg after 1 min	
Kim et al, 2010 [26]	5	Propofol 0	Remifentanyl: TCI 4 ng/mL (10 s)	ASA I/II aged 18-70 y
		Propofol 3 µg/mL		
		Propofol 4.5 µg/mL		
		Propofol 6 µg/mL		
Sedighinejad et al, 2013 [39]	5	Propofol 20 mg	Fentanyl: 4 µg/kg (30 s) after 1 min	ASA I/II aged 25-60 y
		Saline		
		Intralipid	Fentanyl: 2.5 µg/kg (<2 s)	ASA I/II aged 25-60 y
		Propofol 1 mg/kg		
Tang et al, 2010 [24]	6	Propofol 1.5 mg/kg		
		Propofol 2 mg/kg		
		Saline		

Table 1 (continued)

Study	Jadad score	Intervention	Opioid	Patients
Liu et al, 2015 [18]			Sufentanyl: 0.5 µg/kg (>3 s) after 2 min	ASA I/II aged 18-70 y
Xu et al, 2015 [7]	6	Dezocine 0.1 mg/kg		
		Saline	Fentanyl: 3 µg/kg (<5 s)	ASA I/II aged 20-70 y
		Dezocine 0.025 mg/kg		
		Dezocine 0.05 mg/kg		
Sun et al, 2011 [19]	5	Dezocine 0.1 mg/kg		
		Saline	Fentanyl: 5 µg/kg (<2 s) after 10 min	ASA I/II aged 20-60 y
		Saline	Remifentanyl: TCI 5 ng/mL, after 5 min	ASA I aged 18-70 y
		Dexamethasone 10 mg	Fentanyl: 2 µg/kg (>2 s) after 60 min	ASA I
Elmenesy et al, 2011 [43]	6	Dextromethorphan 60 mg		
		Water		
		Dextromethorphan 40 mg	Fentanyl: 2 µg/kg (<2 s) after 60 min	ASA I aged 18-60 y
		Placebo		
An et al, 2015 [31]	7	Saline	Sufentanyl: 1 µg/kg (<5 s) after 1 min	ASA I/II aged 18-65 y
		MgSO ₄ 30 mg/kg		
		MgSO ₄ 50 mg/kg		
		Saline	Fentanyl: 5 µg/kg (<5 s) after 1 min	ASA I/II aged 18-65 y
Liu et al, 2015 [32]	7	MgSO ₄ 30 mg/kg		
		MgSO ₄ 50 mg/kg		
		Control	Fentanyl: 2 µg/kg (>5 s) after 15 min	ASA I/II aged 18-60 y
		Salbutamol		
Agarwal et al, 2003 [6]	5	Beclomethasone		
		Chromoglycate		
		Clonidine 2 mg/kg	Fentanyl: 2 µg/kg (<2 s) after 2 min	ASA I aged 18-80 y
		Saline		
Elmawgoud et al, 2013 [21]	6	Tramadol 1 mg/kg	Fentanyl: 2 µg/kg (>2 s) after 15 min	ASA I aged 18-50 y
		Saline		
		Saline	Fentanyl: 2 µg/kg (>2 s) after 5 min	ASA I aged 19-63 y
		Pentazocine 0.5 mg/kg		
Lui et al, 1996 [40]	5	Saline	Fentanyl: 5 µg/kg after 10 min	ASA I aged 16-45 y
		Terbutaline 5 mg		
		Atropine 0.01 mg/kg		
Horng et al, 2012 [38]	5	Rocuronium 0.06 mg/kg	Fentanyl: 2 µg/kg (>2 s) after 30 s	ASA I-III aged 18-80 y
		Saline		

ASA = American Society of Anesthesiologists.

2.4. Statistical analyses

We employed Review Manager 5.2 (Cochrane Collaboration) to analyze the results. Dichotomous data were calculated with risk ratios, and continuous data were assessed with weighted mean difference, each with 95% confidence intervals. Heterogeneity was analyzed using the χ^2 test; $I^2 > 50\%$ and $P \leq .05$ indicated significant heterogeneity, and the random-effects model was used. If these criteria were not satisfied, the fixed-effects model was used. With regard to the severity of OIC, all the included studies applied the same standard method. Coughing was graded by cough frequency as none, mild (1-2), moderate (3-4), and severe (≥ 5). To facilitate further analysis, we applied the 4-point rating scale (none = 0; mild = 1; moderate = 2; severe = 3) and used the weighted mean difference as the measure of effect.

3. Results

3.1. Study characteristics

Initially, 204 studies were retrieved from the PubMed, Embase, and CENTRAL databases. After the review of titles and abstracts, 123 articles were excluded. In total, 34 RCTs met our inclusion criteria and were finally included in this study [5–7,10,15–44]. The flow diagram of our literature search and detailed steps of selection are shown in Figure 1. A total of 5843 patients who received pharmacological management were considered to be the intervention group, whereas 4063 patients who received placebo were regarded as the control group. A total of 20 different drugs were researched, and 10 medications were discussed in more than 2 studies. The details of the characteristics and quality scores of all included studies are described in Table 1. The Jadad scale scores of all RCTs were 4 to 7 (high quality).

3.2. The incidence of OIC

Pharmacological interventions administered to prevent OIC were as follows: lidocaine, ketamine, dexmedetomidine, priming of fentanyl, propofol, dezocine, dexamethasone, dextromethorphan, MgSO_4 , salbutamol, clonidine, tramadol, pentazocine, rocuronium, midazolam, atropine, terbutaline, sodium chromoglycate, beclomethasone, and ephedrine. Ten regimens were studied just in 1 trial, including clonidine, tramadol, pentazocine, rocuronium, midazolam, atropine, terbutaline, sodium chromoglycate, beclomethasone, and ephedrine. In consideration of the undue weight for these regimens, we made an arbitrary decision to include regimens tested in at least 2 trials. The frequently studied drugs were lidocaine, ketamine, dexmedetomidine, priming of fentanyl, propofol, dezocine, dexamethasone, dextromethorphan, and MgSO_4 , and all of them were significantly more effective than the control. From these data, we found that salbutamol, tramadol, midazolam, and atropine were ineffective in preventing OIC (Table 2; Figs. 2-5).

3.3. The severity of OIC

Among the most frequently studied regimens, 9 medications, including lidocaine, ketamine, dexmedetomidine, priming of fentanyl, propofol, dezocine, dexamethasone, dextromethorphan, and MgSO_4 , produced a decreased severity of OIC. Similarly, salbutamol, tramadol, midazolam, and atropine were found to be ineffective in decreasing the severity of OIC.

3.4. Adverse effects

Sixteen RCTs reported hemodynamic changes in Systolic blood pressure, Diastolic blood pressure, Mean blood pressure, Heart rate, and Pulse oxygen saturation. We did not

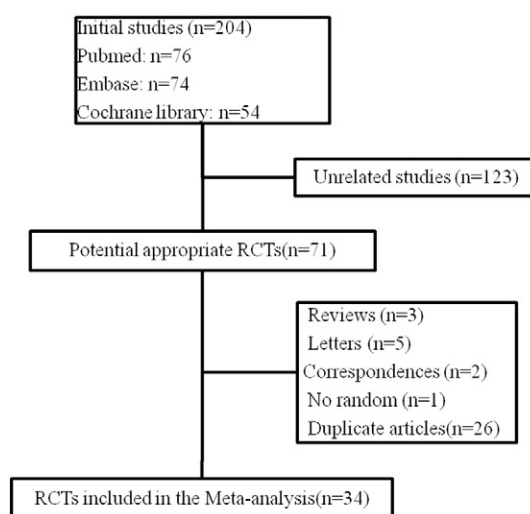


Fig. 1 The flowchart of the meta-analysis.

Table 2 The results of interventions on incidence and severity of OIC

Medicine	No. of RCT	Cough/total patients		Relative risk (95% CI)	Severity of OIC Mean difference
		Intervention	control		
Lidocaine	6	140/949	250/791	0.47 (0.39, 0.56)	-0.25 (-0.35, -0.14)
Ketamine	5	57/566	175/564	0.33 (0.25, 0.43)	-0.51 (-0.72, -0.31)
Dexmedetomidine	4	128/590	175/370	0.49 (0.36, 0.66)	-0.39 (-0.51, -0.26)
Fentanyl	4	210/1400	154/550	0.48 (0.34, 0.65)	-0.28 (-0.44, -0.13)
Propofol	4	82/477	98/217	0.40 (0.24, 0.67)	-0.54 (-0.83, -0.26)
Dezocine	3	14/545	141/345	0.03 (0.00, 0.46)	-0.53 (-0.70, -0.36)
Dexamethasone	2	10/92	35/98	0.30 (0.16, 0.56)	-0.47 (-0.67, -0.27)
Dextromethorphan	2	10/182	102/182	0.15 (0.03, 0.86)	-0.78 (-1.55, -0.01)
MgSO ₄	2	22/184	43/93	0.26 (0.16, 0.41)	-0.70 (-0.93, -0.47)
Salbutamol	2	31/129	38/129	0.55 (0.10, 3.01)	0.29 (0.00, 0.58)
Clonidine	1	26/150	58/150	0.45 (0.30, 0.67)	-0.41 (-0.59, -0.23)
Tramadol	1	10/50	19/50	0.53 (0.27, 1.02)	-0.32 (-0.61, -0.03)
Pentazocine	1	4/92	21/93	0.19 (0.07, 0.54)	-0.28 (-0.43, -0.43)
Rocuronium	1	11/130	30/130	0.37 (0.19, 0.70)	-0.22 (-0.36, -0.08)
Midazolam	1	70/110	45/110	1.56 (1.19, 2.03)	0.70 (0.39, 1.01)
Atropine	1	16/35	13/30	1.05 (0.61, 1.82)	0.08 (-0.54, 0.70)
Terbutaline	1	1/34	13/30	0.07 (0.01, 0.49)	-1.00 (-1.45, -0.55)
Chromoglycate	1	2/50	14/50	0.14 (0.03, 0.60)	-0.42 (-0.66, -0.18)
Beclomethasone	1	0/50	14/50	0.03 (0.00, 0.56)	
Ephedrine	1	6/28	20/31	0.33 (0.16, 0.71)	-0.46 (-0.74, -0.18)

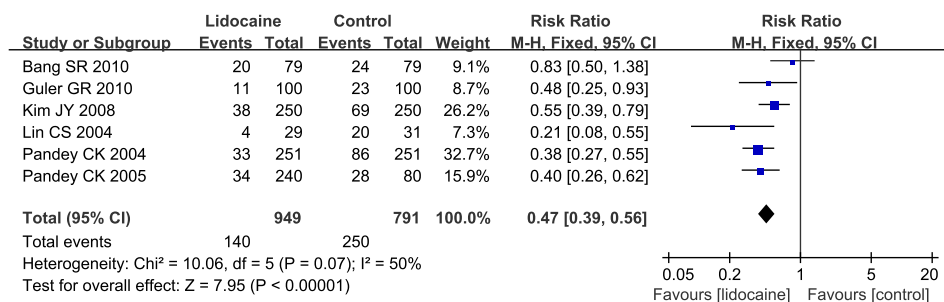
quantitatively analyze them because the assessment methods were not uniform. However, we could conclude that there was no statistically significant difference between the intervention and control groups. There were no other severe adverse effects reported in all studied articles.

4. Discussion

This study concluded that lidocaine, ketamine, dexmedetomidine, priming of fentanyl, propofol, dezocine, dexamethasone, dextromethorphan, MgSO₄, clonidine, pentazocine, rocuronium, terbutaline, sodium chromoglycate, beclomethasone, and ephedrine were all effective in reducing the incidence and severity of OIC. However salbutamol, tramadol, midazolam, and atropine were not effective.

The mechanism of OIC is still unclear, but many factors are associated with it. Those factors can be divided into 2 aspects: the patients' individual physical conditions (age, sex, weight, smoking, and disease history) and the usage of opioids (drug category, dosage, concentration, administration order, injection rate, and site) [2,19,41,45].

Various mechanisms have been presented to explain the occurrence of OIC. Opioids may inhibit central sympathetic outflow and, in turn, activate the vagus nerve. The enhancement of the parasympathetic nervous system is suggested to cause cough and bronchoconstriction [6,40,46]. The pulmonary chemoreflex may be another likely mechanism, as it is mediated by rapidly adapting receptors (irritant receptors) or vagal C-fiber receptors (juxtacapillary receptors) close to pulmonary vessels [47,48]. Moreover, it has been identified that opioid receptors exist in the upper pulmonary mucosa, and opioid-induced tracheal and bronchial smooth muscle

**Fig. 2** Forest plot of the meta-analysis of lidocaine on the incidence of OIC.

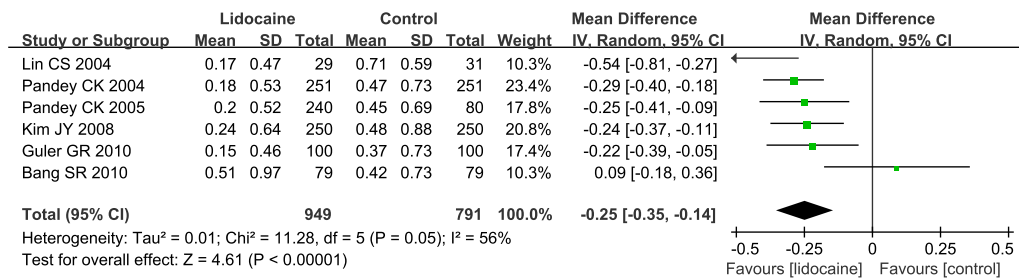


Fig. 3 Forest plot of the meta-analysis of lidocaine on the severity of OIC.

constriction may trigger the cough reflex [49,50]. Furthermore, histamine release from lung mast cells may also contribute to cough. In addition, opioid-induced muscle rigidity, which can cause sudden adduction of the vocal cords or supraglottic obstruction by soft tissue, may be a possible mechanism [11,49].

In this analysis, we found that lidocaine was effective in preventing OIC. The suggested mechanism was that lidocaine inhibits both mechanically and chemically induced airway reflexes by suppressing brain stem function or even anesthetizing peripheral cough receptors distributed in the trachea and hypopharynx [1,51]. Although lidocaine was widely used due to its effectiveness, convenience, low cost, and lower risk of adverse effects, we should also pay attention to the fact that a high dose of lidocaine has an arrhythmogenic effect and may augment cardiovascular depression during anesthesia induction [52].

Ketamine is an *N*-methyl-D-aspartate (NMDA) receptor antagonist; NMDA receptors are widely present in the larynx, lung, and airways. The activation of NMDA receptors on airway smooth muscle would contribute to airway constriction [53,54]. Ketamine may suppress cough by means an antagonistic effect on NMDA receptor. Currently, ketamine is rarely used in general anesthesia due to its adverse effect of hallucinations and the elevation of intracranial pressure, intraocular pressure, and blood pressure.

Dexmedetomidine is a highly selective α_2 -adrenergic agonist. The effect of dexmedetomidine in preventing OIC may be due to the ability of α_2 -adrenergic agonists to reverse muscular rigidity or the relaxation of histamine-induced tracheal

smooth muscle contraction [55,56]. However, the adverse effects of dexmedetomidine include bradycardia and hypotension [57].

Propofol is commonly used in general anesthesia and is considered to have bronchodilation effect. The mechanism of propofol in prevention of OIC may be associated with depression of the NMDA receptor or the direct inhibition of bronchoconstriction [58,59]. However, a high dose of propofol is frequently related to hypotension.

A small priming dose of fentanyl may act on neurotransmitter mechanisms [23]. Dezocine is a mixed agonist-antagonist opioid: it activates κ receptors and antagonizes opioid activated μ receptors [7]. Dexamethasone may activate neutral endopeptidase and, in turn, depress the airway reactivity through airway epithelial cells [60]. The inhibitory action on cholinergic neuromuscular transmission that results in bronchodilation and attenuation of calcium-induced muscle contraction may contribute to the possible effect of magnesium sulfate on OIC [31]. Dextromethorphan may work by antagonizing NMDA receptors in the same way as ketamine [43]. Although various medications can prevent OIC, they are associated with side effects.

A previous meta-analysis reported on the pharmacological and nonpharmacological prevention of fentanyl-induced cough [46]. However, it only included 22 studies. In comparison, we added 12 RCTs to this systematic review, and we analyzed the severity of OIC.

There were some limitations to this study. The number of eligible studies for each drug was small, so the publication bias (tested by funnel plots) was not analyzed. Moreover, some

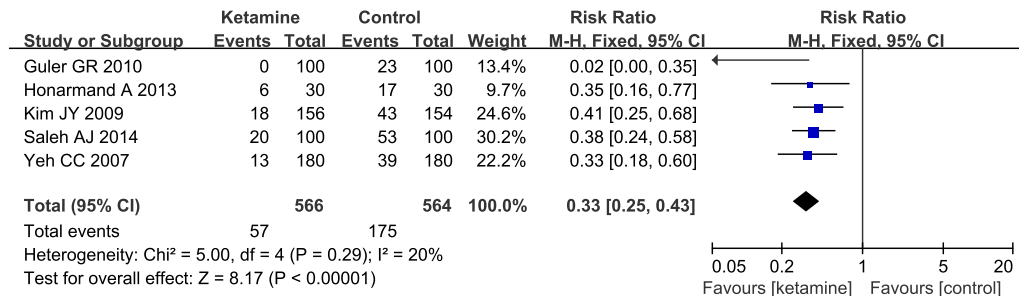


Fig. 4 Forest plot of the meta-analysis of ketamine on the incidence of OIC.

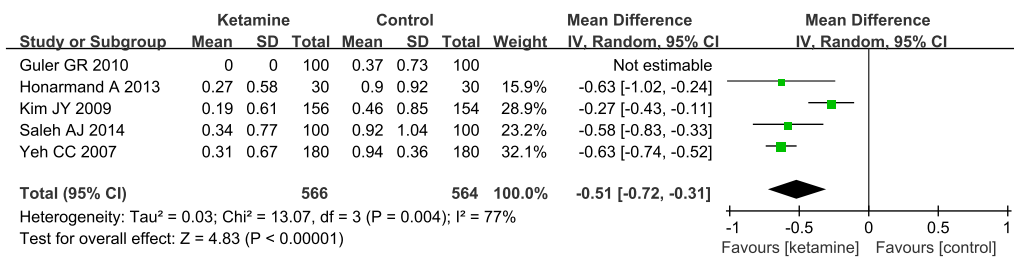


Fig. 5 Forest plot of the meta-analysis of ketamine on the severity of OIC.

interventions had considerable heterogeneity: considering that many factors were associated with the results of OIC, we used the random-effects model.

5. Conclusion

In summary, this study suggested that lidocaine, ketamine, dexmedetomidine, fentanyl, propofol, and dezocine were effective in suppressing OIC. Further studies are needed to confirm the true efficacy of other interventions.

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